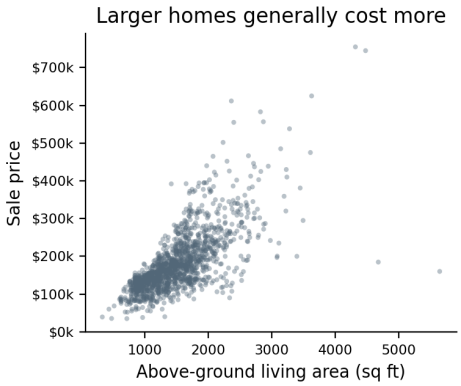
# House prices: what stands out?

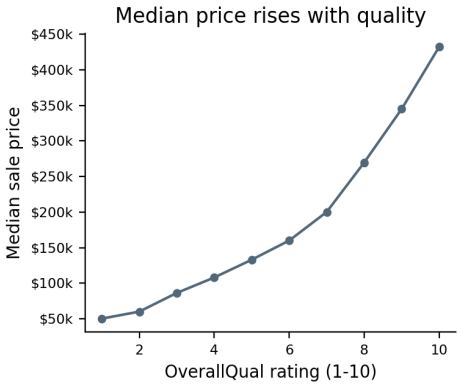
Somtochukwu Kingsley Ekeh | Assignment 2 - Part 3 | Exploratory data analysis

**Scope and data.** The supplied training file contains 1,460 homes and 81 columns, including SalePrice. All prices are present and positive; there are no duplicate rows or IDs. I examine price distributions, numerical correlations, quality ratings, neighbourhood medians, and extreme observations. No prediction model is fitted.

## Price distribution and visual patterns




**Prices are right-skewed.** Mean: $180,921.20; median: $163,000; SD: $79,442.50. The middle 50% spans $129,975-$214,000 (IQR $84,025), with a full range of $34,900-$755,000. Skewness drops from 1.883 to 0.121 after taking natural logs. Log price is more symmetric, but this does not establish normality.

## Variables associated with SalePrice

| Variable | Pearson r | Spearman rho | Available pairs |
|---|---|---|---|
| OverallQual | 0.791 | 0.810 | 1460 |
| GrLivArea | 0.709 | 0.731 | 1460 |
| GarageCars | 0.640 | 0.691 | 1460 |
| GarageArea | 0.623 | 0.649 | 1460 |
| TotalBsmtSF | 0.614 | 0.603 | 1460 |

**Quality and space show the strongest observed associations.** OverallQual is ordinal, so Spearman correlation is included. Median prices are $133,000 at quality 5, $200,141 at 7, and $269,750 at 8. Ratings 1 and 2 have just 2 and 3 homes. Garage capacity and garage area also overlap, so their correlations are not separate independent effects.

**Neighbourhood medians differ.** MeadowV has a median of $88,000 (n = 17), versus $315,000 in NridgHt (n = 77). Small groups require care; Blueste has only 2 homes. Size, quality, and age may explain part of these descriptive location differences.

**Missing codes and extremes.** NA codes include PoolQC (1,453), MiscFeature (1,406), Alley (1,369), and LotFrontage (259). Depending on the field, a code may indicate feature absence or an unknown measurement. The raw CSV and literal None category are preserved; no blanket imputation is used. Pearson correlations use available pairs. The 1.5 x IQR price rule flags 61 homes (4.18%) above $340,037.50; these are retained. Living-area r changes from 0.709 to 0.721 when homes above 4,000 sq ft are excluded only as a sensitivity check.

**Conclusion and limits.** Prices have a long upper tail, and quality, living area, garage attributes, and neighbourhood show clear descriptive associations. Correlation does not show causation or an independent price effect. Category coding, missing/absent features, unequal group sizes, and extreme properties limit interpretation. These training records alone do not establish future predictive accuracy or results for another market or period.

Source: supplied train(2).csv, House Prices training data. Notebook, tables, plots, raw data, README, and reproducible script are included in the organized project.